

SEMESTER 5

INFORMATION TECHNOLOGY

SEMESTER S5

MACHINE LEARNING

Course Code	PCITT501	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PBITT404 Data Science	Course Type	Theory

Course Objectives:

1. Gain a comprehensive core machine learning concepts, pre-process and analyze data, build and assess supervised learning models, and interpret their performance.
2. Acquire the knowledge of implementing and fine-tuning advanced machine learning models, including SVMs and neural networks, and effectively use unsupervised learning, reinforcement learning, and deep learning methods to solve complex problems.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Overview of Machine Learning - Introduction to Machine Learning, Examples of Machine Learning Applications - Learning Associations, Classification, Regression, Unsupervised Learning, Reinforcement Learning. Parametric Methods - Maximum Likelihood Estimation- Bernoulli Density, Multinomial Density, Gaussian (Normal) Density, Evaluating an Estimator: Bias and Variance, The Bayes' Estimator, Parametric Classification, Regression, Tuning Model Complexity: Bias/Variance Trade-off. Dimensionality Reduction – Principal Component Analysis, Linear Discriminant Analysis.	10
2	Supervised Learning - Linear Regression - Least squares, Gradient Descent, Assessing Performance: Error Metrics, Overfitting, Linear Classifiers - Logistic regression, Naïve Bayes', Measuring Classifier Performance – Confusion Matrix, Cross Validation. Bootstrapping, Model Selection Procedures, Regularization Techniques – LASSO, RIDGE. Learning with Trees - Decision Trees, Types- Classification and Regression	12

	Trees, Feature Selection Measures - Entropy, Information Gain, ID3 Algorithm, Issues in Decision tree Learning, Ensemble Learning – Boosting – Bagging.	
3	Support Vector Machine – Optimal Separating hyper plane, Soft-margin & Hard- margin, Kernel trick, Kernel functions. Neural Networks – The Perceptron Algorithm, Activation Functions, Feed Forward Network, Back Propagation Algorithm.	10
4	Unsupervised Learning - Clustering Methods - K-means, Expectation-Maximization Algorithm, Density based clustering. Reinforcement Learning - Monte Carlo, TD Learning, Q Learning. Basics of Deep Learning - Convolutional Neural Networks – Convolution, Pooling Layers, Hyperparameters, Transformers.	12

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE):

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• 2 questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. • Each question carries 9 marks. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain basic machine learning concepts, parametric methods for estimation and dimensionality reduction techniques to improve models.	K2
CO2	Use supervised learning methods like linear regression and decision trees, evaluate model performance and ensemble techniques.	K3
CO3	Use Support Vector Machines and Neural Networks, including understanding margins, kernel functions, and the basics of the perceptron algorithm and back propagation.	K3
CO4	Apply clustering methods, reinforcement learning techniques, and basic deep learning concepts like convolutional neural networks and transformers.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	2	3		2	2	-	-	-	-	-	2
CO2	3	3	3		1	2	-	-	-	-	-	2
CO3	3	2	2		2	1	-	-	-	-	-	3
CO4	2	2	2		2	-	-	-	-	-	-	3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to Machine Learning	Ethem Alpaydin	MIT Press	Third Edition, 2014
2	Machine Learning	Tom M Mitchell	McGraw Hill Education	First Edition, 2013
3	Reinforcement Learning – An Introduction	Richard S. Sutton, Andrew G. Barto, Francis Bach	MIT Press	Second Edition, 2018
4	Deep Learning	Ian Goodfellow, Yoshua Bengio and Aaron Courville	MIT Press	1st Edition, 2016.

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Data Mining Concepts & Techniques	Jiawei Han, Micheline Kamber and Jian Pei.	Morgan Kaufmann	3 rd Edition, 2012
2	An Introduction to Statistical Learning: with Applications in R	Gareth James, Daniela Witten, Trevor Hastie , Robert Tibshirani	Springer	2 nd Edition, 2021
3	Neural Networks for Pattern Recognition	Christopher Bishop	Oxford University Press	1995
4	Neural Networks and Deep Learning: A Textbook	Charu C. Aggarwal	Springer	2019

Video Links (NPTEL, SWAYAM...)	
No.	Link ID
1	https://nptel.ac.in/courses/106106139
2	https://archive.nptel.ac.in/courses/106/106/106106184
3	https://nptel.ac.in/courses/106106198

SEMESTER S5

ALGORITHM ANALYSIS AND DESIGN

Course Code	PEITT502	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:1:0:0	ESE Marks	60
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PCITT303 - Data structures	Course Type	Theory

Course Objectives:

1. Understand and Apply Fundamental Algorithmic Concepts
2. Master Algorithm Design Paradigms and Complexity Analysis

SYLLABUS

Module No.	Syllabus Description	Contact Hours (44 Hrs)
1	Algorithms – Properties of a Good algorithm, Characteristics, Criteria for Analysing Algorithms. Time and Space Complexity - Best, Worst and Average Case Complexities. Asymptotic Notations and their properties. Time and Space Complexity Calculation of simple algorithms. Analysis of Recursive Algorithms: Recurrence Equations, Solution of Recurrence Equations – Iteration Method, Recursion Tree Method, Substitution method and Master's Theorem (Proof not required). Amortized Complexity – aggregate analysis, Cost-Accounting and Potential Methods.	11
2	Balanced Search Trees - AVL Trees (Insertion and deletion operations with all rotations in detail, algorithms not expected); Red Black Trees – Properties Disjoint Sets- Disjoint set operations, Union and find algorithms, Analysis of union by rank with path compression, Connected components of a Graph. Graphs – Representations, Traversals – BFS, DFS and their analysis, Strongly Connected Components, Topological Sorting.	11

	Divide and Conquer – Control Abstraction, Finding Minimum and Maximum, Quicksort, Merge Sort, Strassen’s Matrix Multiplication, analysis	
3	Greedy Algorithms: Control Abstraction, Fractional Knapsack, Minimum Cost Spanning Tree – Kruskal’s and Prim’s, analysis, Job Sequencing with deadlines Dynamic Programming: Control Abstraction, Optimality Principle, DP Solution for Travelling Salesman Problem and 0/1 Knapsack Problem, Backtracking: Control Abstraction, N – Queens Problem, algorithm.	11
4	Branch and Bound: Control Abstraction, FIFO, LIFO and LC Branch and Bound, 15 – Puzzle Problem String Matching – KMP Algorithm, Rabin-Karp Algorithm. Tractable and Intractable Problems, Complexity Classes – P, NP, NP- Hard and NP-Complete Classes. Approximation algorithms- Planar Graph Coloring, Vertex Cover	11

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Micro project	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE):

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain asymptotic notations used in the performance analysis of algorithms, to solve recurrence equations and study the amortized analysis.	K2
CO2	Apply divide and conquer strategy to solve practical problems efficiently	K3
CO3	Apply greedy and dynamic programming techniques in algorithm design	K3
CO4	Apply backtracking and branch and bound techniques in algorithm design.	K3
CO5	Explain Balanced Search Trees, Graph Traversals, and sophisticated algorithms such as string matching and approximation algorithms	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2	3	-	-	-	-	1	1	-	2
CO2	3	3	3	3	2	-	-	-	1	1	-	2
CO3	3	3	3	3	2	-	-	-	1	1	-	2
CO4	3	3	3	3	2	-	-	-	1	1	-	2
CO5	3	3	3	3	-	-	-	-	1	1	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to Algorithms	T.H.Cormen, C.E.Leiserson, R.L.Rivest, C. Stein	Prentice-Hall India	2nd Edition (2001)
2	Fundamentals of Computer Algorithms	Ellis Horowitz, Sartaj Sahni, Sanguthevar Rajasekaran	Orient Longman Universities Press	2nd Edition (2008)
3	Computer Algorithms, Introduction to Design and Analysis	Sara Baase and Allen Van Gelder	Pearson Education	3rd Edition (2009)

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Algorithm Design	Jon Kleinberg and Éva Tardos	Pearson	1st Edition 2005
2	Algorithms	Robert Sedgewick and Kevin Wayne	Addison-Wesley	4th Edition 2011
3	Computers and Intractability: A Guide to the Theory of NP-Completeness	Michael R. Garey and David S. Johnson	W. H. Freeman	1st Edition 1979

Video Links (NPTEL, SWAYAM...)	
No.	Link ID
1	https://onlinecourses.nptel.ac.in/noc24_cs79/

SEMESTER S5
SOFTWARE ENGINEERING

Course Code	PCITT503	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	Theory

Course Objectives:

1. To understand the fundamental concepts of Software Engineering, Software Development Life Cycle (SDLC) and Software Process Models.
2. To develop proficiency in Object-Oriented Modeling and Design.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	The Software Process: Software Engineering, Software Process, Software Engineering Practice, Phases of Software Development Life Cycle (SDLC)- Requirements, Design, Implementation, Testing, Deployment and Maintenance, Software Process models- Prescriptive and Specialized Process Models, Characteristics of Software Engineer, Software Team, Team Structure, Software Engineering Ethics.	9
2	Requirements Engineering: Functional and Non-Functional Requirements, Requirement Engineering Process, Requirements elicitation, Requirements Specification, SRS Template, Requirements Validation, Requirements Change, Requirement Analysis, Case Study.	9
3	Object Oriented Modeling: Object Oriented Development, Modeling, Abstraction, Objects and Classes, Link and Association Concepts, Generalization and Inheritance, N-ary Association, Abstract classes, Aggregation, Packages, State Modeling, Interaction Modeling, Structural Modeling, Behavioral Modeling.	9

4	Object Oriented Design: Design Process, Design Concepts- Cohesion, Coupling, Design Model, System Design, Class Design, Architectural Design- Architectural design decisions, Architectural Views, Architectural Patterns and Application architectures, Component-level Design, Design and Implementation- Object-oriented design using the UML, Open-source development.	9
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE):

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Discuss the software process and lifecycle models of software engineering, including the roles of software engineers in various phases of development.	K2
CO2	Apply various techniques to elicit requirements from stakeholders, create detailed requirement specifications, and ensure that requirements are well-defined and traceable throughout the software development process.	K3
CO3	Apply object-oriented modeling techniques to design and document software systems.	K3
CO4	Explain the fundamental principles of software design, including design process and architectural patterns.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	-	-	-	2	-	-	2	2	-	2	2
CO2	2	2	3	2	2	2	-	-	2	2	2	2
CO3	2	2	3	2	3	2	-	-	2	2	2	2
CO4	2	2	3	2	3	2	-	-	2	2	3	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Software Engineering: A Practitioner's Approach	Roger S Pressman, Bruce R Maxim	Tata McGraw Hill	8th edition, 2019
2	Software Engineering	Ian Somerville	Pearson	10th edition, 2017
3	Object-oriented modeling and design with UML	Michael R Blaha, James R Rumburg	Pearson	2nd edition, 2007

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Fundamentals of Software Engineering	Rajib Mall	PHI	5th edition, 2018
2	Software Engineering: Theory and Practice	Shari Pfleeger, Joanne Atlee	Pearson	4th edition, 2009
3	Applying UML and Patterns: An Introduction to Object-Oriented Analysis and Design and Iterative Development	Craig Larman	PHI	3rd edition, 2004
4	Design Patterns: Elements of Reusable object-oriented software	Erich Gamma	Pearson	1st edition, 2015

Video Links (NPTEL, SWAYAM...)	
No.	Link ID
1	https://onlinecourses.nptel.ac.in/noc24_cs119/preview

SEMESTER S5

WEB APPLICATION DEVELOPMENT

Course Code	PBITT504	CIE Marks	60
Teaching Hours/Week (L: T:P: R)	3:0:0:1	ESE Marks	40
Credits	4	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	GYEST203: Foundations of Computing: From Hardware Essentials to Web Design and GYEST204: Programming in C	Course Type	Theory

Course Objectives:

1. Students will be able to gain a comprehensive understanding of javascript, node.js and mongodb to develop web applications.
2. Students will be able to develop web application based solutions to real world problems using modern web application development tools.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Javascript: How javascript makes web pages more interactive, Examples of javascript in the browser, How html, css & javascript fit together, creating a basic javascript, placing the script in the page, where to place your scripts, <i>basic Javascript instructions:</i> statements, comments, variables, data types, arrays, expressions, operators; <i>functions methods and objects:</i> function, anonymous function, variable scope, object, this keyword, arrays of objects and objects in arrays built-in objects, browser object model, document object model, <i>Global objects:</i> string, number, math, date DOM: Document Object Model (DOM), working with DOM tree, accessing elements, nodelists, <i>selecting elements:</i> using class attribute, tag name, css selectors; repeating actions for an entire nodelist, looping through a nodelist, traversing the DOM, previous & next sibling, first & last child, textcontent, adding or removing html content, update text and markup, adding/removing elements, creating attributes and changing their values, removing attributes Event handling: different event types, three ways to bind an event to an element, using DOM event handlers, using event listeners, using parameters with event listeners, the event object, event bubbling, user interface events	9
2	ECMA Script: ECMA Script versions, ES5 Features, ES6 introduction, var declarations and hoisting, let declaration, constant declaration, function with default parameter values, default parameter expressions, unnamed parameters, the spread operator, arrow functions, object destructuring, array destructuring, sets and maps, Array.find, Array.findIndex, template strings,	10

	javascript classes, callbacks, promises, async/await AJAX: What is Ajax?, why use ajax?, How ajax works?, Handling ajax request and response, <i>data formats</i>: XML, JSON; Working with JSON data, Loading HTML with Ajax, Loading JSON with ajax, working with data from other servers JQUERY: A basic jquery example, Why use JQuery?, finding elements, jquery selection, getting element content, updating elements, changing content, inserting elements, adding new content, getting and setting attributes, getting and setting css properties, using .each(), events, event object, effects, animating css properties, using animation, traversing the dom, working with forms, jquery and ajax	
3	Node.js: What is Node.js?, installing Node.js, creating first server, request and response, responding with HTML, installing custom packages with NPM, introduction to express, handling request with express, serving other html files, serving static files with express, automatic server restart with nodemon, public folder for serving static files, creating page routes, templating engines MongoDB: Architecture of MongoDB, installing MongoDB, mongoose, defining a model, CRUD operations with Mongoose models, reading data from MongoDB using Mongoose, updating record, deleting a single record, saving to the database, dynamic data with templating engines, adding fields to the schema, uploading image with express, saving uploaded images to database	10
4	Refactoring to MVC: Express middleware, custom middleware, registering validation middleware, refactoring to MVC, refactoring validation layer, mongoose validation, password encryption, user login process, user authentication with express sessions, authentication middleware, user logout, handling non-existent routes, displaying validation errors in templates, duplicate entry error, flushing error messages from session, customising error message, persist request data on form, adding a WYSIWYG editor Deployment: setting up MongoDB atlas, deploying web apps on Heroku, deployment	7

Suggestions on Project Topics:

No.	Project Topics
1	Countdown clock (javascript)
2	To do list (javascript)
3	Rock paper scissors game (javascript)
4	Shopping cart page (javascript)
5	Calculator app (javascript)

6	Memory game (javascript)
7	Movie app (Node.js, Express, MongoDB)
8	Chat app (Node.js, Express, MongoDB)
9	Hotel booking app (Node.js, Express, MongoDB)
10	Library app (Node.js, Express, MongoDB)
11	Job search app (Node.js, Express, MongoDB)
12	E-commerce website (Node.js, Express, MongoDB)

Course Assessment Method
(CIE: 60 marks, ESE: 40 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Project	Internal Examination-1 (Written)	Internal Examination-2 (Written)	Total
5	30	12.5	12.5	60

End Semester Examination Marks (ESE):

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 2 marks (8x2 =16 marks)	- Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 2 sub divisions. - Each question carries 6 marks. (4x6 = 24 marks)	40

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Apply the fundamental concepts of javascript to add user interaction in the website.	K3
CO2	Compare javascript, jquery in terms of performance and ease of use.	K2
CO3	Build web applications with Node.js, express and MongoDB.	K3
CO4	Develop and deploy web applications with Node.js, express, MongoDB adhering to MVC architecture.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2	3	-	2	-	-	-	2	-	-	2
CO2	2	2	3	3	3	-	-	-	3	-	2	2
CO3	3	2	3	3	3	-	-	-	3	-	2	3
CO4	3	2	3	3	3	-	-	-	3	-	2	3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Javascript and JQuery: Interactive front end development	Jon Ducket	Wiley	1 st Edition 2013
2	Understanding ECMAScript6	Nicholas C Zakas	No starch press	1 st Edition 2016
3	Beginning Node.js, Express and mongoDB development	Greg Lim	Greg Lim	1 st Edition 2019

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	HTML & CSS : Design and build websites	Jon Duckett	Wiley	1 st Edition 2011
2	Internet & World Wide Web how to programme	Paul Deitel, Harvey Deitel, Abbey Deitel	Deitel	5 th Edition 2013
3	Web development with Node & Express	Ethan Brown	O'Reilly	1 st Edition 2014

Video Links (NPTEL, SWAYAM...)	
No.	Link ID
1	https://onlinecourses.swayam2.ac.in/nou24_cs09/preview

Project Based Learning - Course Elements			
L: Lecture (3 Hrs.)	R: Project (1 Hr.), 2 Faculty Members		
	Tutorial	Practical	Presentation
Lecture delivery	Project identification	Simulation/ Laboratory Work/ Workshops	Presentation (Progress and Final Presentations)
Group discussion	Project Analysis	Data Collection	Evaluation
Question answer Sessions/ Brainstorming Sessions	Analytical thinking and self-learning	Testing	Project Milestone Reviews, Feedback, Project reformation (If required)
Guest Speakers (Industry Experts)	Case Study/ Field Survey Report	Prototyping	Poster Presentation/ Video Presentation: Students present their results in a 2 to 5 minutes video

Assessment and Evaluation for Project Activity		
Sl. No	Evaluation for	Allotted Marks
1	Project Planning and Proposal	5
2	Contribution in Progress Presentations and Question Answer Sessions	4
3	Involvement in the project work and Team Work	3
4	Execution and Implementation	10
5	Final Presentations	5
6	Project Quality, Innovation and Creativity	3
Total		30

1. Project Planning and Proposal (5 Marks)

- Clarity and feasibility of the project plan
- Research and background understanding
- Defined objectives and methodology

2. Contribution in Progress Presentation and Question Answer Sessions (4 Marks)

- Individual contribution to the presentation
- Effectiveness in answering questions and handling feedback

3. Involvement in the Project Work and Team Work (3 Marks)

- Active participation and individual contribution
- Teamwork and collaboration

4. Execution and Implementation (10 Marks)

- Adherence to the project timeline and milestones
- Application of theoretical knowledge and problem-solving
- Final Result

5. Final Presentation (5 Marks)

- Quality and clarity of the overall presentation
- Individual contribution to the presentation
- Effectiveness in answering questions

6. Project Quality, Innovation, and Creativity (3 Marks)

- Overall quality and technical excellence of the project
- Innovation and originality in the project
- Creativity in solutions and approaches

SEMESTER S5
SOFT COMPUTING

Course Code	PEITT521	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PCITT303 Data structures	Course Type	Theory

Course Objectives:

1. To provide students with an understanding of computing techniques that handle imprecision, uncertainty, and approximation.
2. To equip students with both theoretical knowledge and practical skills in soft computing, preparing them for advanced studies or professional work in the field.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Introduction to Soft Computing and Neural Networks: Evolution of Computing, Soft Computing Definition, Comparison with Hard Computing, Soft Computing Basic Components, Scope and Applications, Structure of Artificial Neuron and Biological Neuron, Activation functions, McCulloch-Pitts Neuron, Single-layer Perceptron, Multi-layer Feedforward Networks, Learning Logic GATE Functions, Backpropagation Neural Networks.	9
2	Fuzzy Logic and Fuzzy Sets: Fuzzy Logic, Classical Sets, Fuzzy Sets, Fuzzy Relations, Operations on Fuzzy Relations, Fuzzy Membership Functions, Fuzzification, Methods of Membership Value Assignments, Defuzzification Methods.	9
3	Fuzzy Rule Base and Approximate Reasoning: Fuzzy Rules- Formation, Decomposition and Aggregation, Fuzzy Reasoning, Fuzzy Inference System- Mamdani Fuzzy Inference System and Takagi Sugeno Fuzzy Model, Comparison between Mamdani and Sugeno Method, Fuzzy Decision Making.	9

4	Genetic Algorithm and Hybrid Systems: Basic Version of Genetic Algorithm, Encoding Methods, Selection Mechanisms, Crossover Techniques, Mutation Operators, Hybrid Soft Computing Systems- Neuro Fuzzy Hybrid System, Neuro Genetic Hybrid System and Fuzzy Genetic Hybrid System.	9
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE):

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Describe the role of soft computing techniques and apply various artificial neural networks to learn and approximate logic gate functions.	K3
CO2	Design and implement fuzzy logic systems to handle uncertainty and imprecision and apply these systems to solve engineering problems and decision-making tasks.	K3
CO3	Apply genetic algorithms to solve complex optimization problems, including defining suitable fitness functions and selecting appropriate genetic operators.	K3
CO4	Explain hybrid systems that integrate multiple soft computing techniques to address complex problems.	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2	2	2	2	-	-	2	-	-	2
CO2	3	2	3	3	2	2	-	-	2	-	2	2
CO3	3	3	3	3	3	3	-	-	2	-	2	2
CO4	3	2	2	2	2	2	-	-	2	-	2	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Principles of Soft Computing	S.N. Sivanandam, S.N. Deepa	Wiley India Pvt. Ltd.	2nd Edition, 2011
2	Neural Networks, Fuzzy Logic and Genetic Algorithm, Synthesis and Applications	S. Rajasekaran, G.A. Vijayalakshmi Pai	PHI Learning Pvt. Ltd.	2017
3	Fuzzy Sets and Fuzzy Logic-Theory and Applications	George J. Klir, Bo Yuan	Prentice Hall	1996
4	Neuro-Fuzzy and Soft Computing	Jyh-Shing Roger Jang, Chuen-Tsai Sun, Eiji Mizutani	Prentice-Hall	2002

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Neural Networks Algorithms, Applications, and Programming Techniques	James A. Freeman, David M. Skapura	Addison Wesley	2003
2	Introduction to Artificial Neural Systems	Jacek M. Zurada	PWS Publishers	1992
3	An Introduction to Genetic Algorithm	Mitchell Melanie	Prentice Hall	1998
4	Genetic Algorithms in Search, Optimization and Machine Learning	David E. Goldberg	Addison Wesley	1997

Video Links (NPTEL, SWAYAM...)	
No.	Link ID
1	https://archive.nptel.ac.in/courses/106/105/106105173/

SEMESTER S5

INTERNETWORKING USING TCP/IP

Course Code	PEITT522	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PCITT402 Computer Networks	Course Type	Theory

Course Objectives:

1. To familiarize with various protocols in TCP/IP suit.
2. To equip students to apply their knowledge in solving issues in handling IP networks
3. To familiarize with the different routing scenarios in unicasting & multicasting in IP networks.
4. To familiarize with label switching & technology and MPLS.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	TCP/IP protocol suite- Comparison between OSI and TCP/IP Protocol Suite Layers in the TCP/IP Protocol Suite, Addressing, IPv4 Addresses - Address Space, Notation, Range of Addresses, Classful Addressing, Classless Addressing (simple problems on subnetting), Special Addresses, NAT, Delivery and Forwarding of IP Packets - Delivery, Forwarding - Forwarding Based on Destination Address, Forwarding with Classful Addressing (without and with subnetting), Forwarding with Classless Addressing, Address Aggregation, Longest Mask Matching, Structure of a Router.	9
2	IPv4 - Datagram, Fragmentation, Options, Checksum, Address Mapping- ARP Protocol, ARP Package, ICMPv4- ICMP encapsulation, Messages, Debugging Tool, ICMP package, Mobile IP - addressing, agents, three phases, inefficiency in mobile IP, Unicast Routing Protocols - RIP, OSPF - Areas, Types of Links, OSPF Packets, Common Header, Link State Update Packet (general format & LSA	9

	general header only), Hello Message, Database Description Message, Link State Request Packet, Link State Acknowledgment Packet), BGP.	
3	Multicasting and Multicast Routing Protocols - Multicasting, Multicast Addresses in IPv4, Delivery of Multicast Packets at Data Link Layer, IGMP - Group Management, IGMP Messages (Membership Query Message Format, Three Formats of Query Messages, Membership Report Message Format), Multicast routing, Multicast routing protocols - MOSPF, DVMRP, PIM.	9
4	TCP - TCP Services, TCP features, TCP Segment format & encapsulation, State transition diagram (TCP connection management), Windows in TCP, TCP timers, TCP options, TCP Package, IPv6 - Notations, Address Space, Three address types, Address Space allocation Algorithm, Auto Configuration, Renumbering, Packet Format (base header and extension headers), Label Switching, Flows, and MPLS- Switching Technology, Flows and Flow Setup, Large Networks, Label Swapping, and Paths, IP Switching Technologies and MPLS, Labels and Label Assignment, MPLS Encapsulation, Label Switching Router (Text Book 2)	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE):

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Apply the IP addressing & IP forwarding concepts in solving addressing problems	K3
CO2	Understand the IP layer unicast routing protocols ICMPv4 and ARP.	K2
CO3	Illustrate the concepts of multicasting and multicast routing protocols.	K2
CO4	Summarize the TCP protocol concepts and TCP package.	K2
CO5	Understand IPv6, Label Switching, Flows, And MPLS	K2

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2	3	2	-	-	-	-	-	-	-	3
CO2	2	2	2	2	-	-	-	-	-	-	-	3
CO3	2	2	-	2	-	-	-	-	-	-	-	3
CO4	2	2	-	2	-	-	-	-	-	-	-	3
CO5	2	2	2	2	-	-	-	-	-	-	-	3

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	TCP/IP Protocol Suite	Behrouz A. Forouzan	McGraw-Hill	Fourth Edition
2	Internetworking With TCP/IP Vol I: Principles, Protocols, and Architecture	Douglas E. Comer	Pearson Education	Sixth Edition

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Computer Networks: A Systems Approach	Larry Peterson	Larry Peterson and Bruce Davie	Computer Networks: A Systems Approach

Video Links (NPTEL, SWAYAM...)	
No.	Link ID
1	https://onlinecourses.nptel.ac.in/noc24_cs69/preview

SEMESTER S5 CLOUD COMPUTING

Course Code	PEITT523	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PCITT402 Computer Networks PCITT403 Operating Systems	Course Type	Theory

Course Objectives:

1. Gain a comprehensive understanding of cloud computing principles, including service and deployment models, and the key characteristics that define cloud computing.
2. Familiarize with the cloud computing reference architecture and understand the significance of virtualization in cloud computing.
3. Explore various cloud software environments and the architectural design of computing and storage clouds.
4. Learn about the security challenges in cloud computing, the role of cryptography, and various architectures to ensure secure data storage and processing in the cloud.
5. Gain proficiency in parallel and distributed programming models and develop practical skills in using major cloud platforms.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Cloud Computing – Introduction. Overview of Computing Paradigms-Grid Computing, Cluster Computing, Distributed Computing, Utility Computing. Cloud Service Models - Software as a Service, Platform as a Service, Infrastructure as a Service. Cloud Deployment Models - Private Cloud, Public Cloud, Community Cloud, Hybrid Cloud. Characteristics of Cloud Computing. Advantages and disadvantages of cloud computing. Cloud Computing Reference Architecture-Cloud Consumer, Cloud Provider, Cloud Auditor, Cloud Broker, Cloud Carrier, IBM Cloud Computing Reference Architecture. Cloud Computing Use Cases - Business Use Cases, Technical Use Cases	9
2	Virtualization- Introduction. Characteristics of Virtualization. Importance of Virtualization. Classification of Virtualization-Network Virtualization, Desktop Virtualization, Application Virtualization, Central Processing Unit Virtualization, Memory Virtualization, Input/Output Devices Virtualization. Virtualization to Private Cloud Services -Five High-Level Steps. Virtualization Architecture-Virtualization Platforms, Hypervisor. Tools and Mechanisms for Implementation of Virtualization. Physical and Virtual Clusters and Resource Management. Importance	10

	of Virtualization in Data Center Automation.	
3	Architectural Design of Compute and Storage Clouds-Data Center Interconnection Networks and Design, Architectural Design of Computing Clouds, Cloud Service Models and Platforms, Resource Management and Design Challenges. Fundamental Cloud Security-Threat agents, Cloud security threats, Operating system security, Infrastructure security- Network Level Security, Host Level Security, Application level security. Secure Distributed Data Storage in Cloud - Cryptography as a Primary Technique to Ensure Security, Gartner's Seven Cloud Computing Security Risks, Need for Data Security. Cloud Computing—Attacks. Cloud Data Security Requirements. Cloud Security—Controls. Cloud Security Architecture. Cloud Data Security.	10
4	Parallel and Distributed Programming Models. MapReduce- MapReduce Programming Model, Logical Data Flow and Control Flow, Case Study: MapReduce Program to Find Inverted Index of Documents. Twister and Iterative MapReduce, Hadoop Library, Apache Spark. Cloud Software Environments-Eucalyptus, OpenNebula, OpenStack, Aneka, CloudSim. Popular Cloud Databases – MongoDB, HBase. Popular Cloud Platforms. Amazon Web Services- Amazon Web Services (AWS)- Elastic Compute Cloud (EC2), Simple Storage Service (Amazon S3), Elastic Block Store (EBS), Amazon CDN services. Google Cloud Platform (GCP)- Compute Engine (GCE), Google App Engine (GAE), Cloud Storage, Gmail, Google Drive. Microsoft Azure- Azure Virtual Machine, Hyper-V, Azure Storage services.	9

**Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)**

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE):

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
2 Questions from each module. - Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	- Each question carries 9 marks. - Two questions will be given from each module, out of which 1 question should be answered. - Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Describe the fundamental concepts of cloud computing, including service and deployment models, and the cloud computing reference architecture.	K2
CO2	Explain the principles of virtualization and its role in enabling cloud services, including different types of virtualization and their applications.	K2
CO3	Identify the architectural components of compute and storage clouds and discuss basic security challenges in cloud computing.	K2
CO4	Implement parallel and distributed programming models using MapReduce and utilize cloud databases like MongoDB and HBase.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	-	-	2	1	-	-	-	-	-	-	2
CO2	3	2	-	2	1	-	-	-	-	-	-	2
CO3	2	1	-	2	2	-	-	-	-	-	-	3
CO4	3	3	2	3	2	-	-	-	-	-	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Cloud Computing & Big Data From the basics to practical use cases	Sudheep E. M., Sarith D. M., Lija M., Tanmay K. P. & Shubham A.	Cengage Learning	1 st Edition, 2024
2	Cloud Computing Concepts, Technology & Architecture	Thomas, E., Zaigham M., & Ricardo P.	Prentice Hall	1 st Edition, 2013
3	Mastering cloud computing: foundations and	Buyya, R., Vecchiola, C., & Selvi, S. T.	Morgan Kaufmann	1 st Edition, 2017

	applications programming			
4	Cloud computing	Bhowmik, S.	Cambridge University Press	1 st Edition, 2017

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Cloud computing: theory and practice	Marinescu, D. C.	Morgan Kaufmann	2 nd Edition, 2017
2	Cloud computing: Principles and paradigms	Buyya, R., Broberg, J., & Goscinski, A. M.	John Wiley & Sons.	1 st Edition, 2011

Video Links (NPTEL, SWAYAM...)	
No.	Link ID
1	https://youtu.be/NzZXz3fJf6o
2	https://youtu.be/fZ3D6HQRWzs
3	https://youtu.be/LcAPj95KeSA
4	https://youtu.be/TOOSVsxElpo

SEMESTER S5

DATA MINING AND WAREHOUSING

Course Code	PEITT524	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PBITT404 Data Science	Course Type	Theory

Course Objectives:

1. To understand the fundamental concepts of data mining and data warehousing.
2. To apply data mining techniques to process and interpret large datasets.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Data mining Introduction –Types of data to be mined - Data mining functionalities - Classification of Data mining system- Data mining task primitives - Major issues in Data mining. Data pre-processing – Need of data pre-processing - Descriptive Data summarization – Data cleaning- Data integration and transformation - Data reduction -Data discretization and Concept Hierarchy Generation.	11
2	Data Warehouse and OLAP Technology: Overview – Multidimensional Data Model- Data Warehouse Architecture - Data Warehouse Implementation – From Data Warehousing to Data Mining.	9
3	Classification Techniques - Decision Trees - Naive Bayes - k-Nearest Neighbors - Support Vector Machines Clustering Techniques - k-Means - Hierarchical Clustering and DBSCAN. Association Rule Mining - Apriori algorithm - FP-Growth Algorithm. Model Evaluation and Validation - Cross-validation, confusion matrix, ROC curves.	10

4	Advanced Data Mining Techniques: Multimedia Data Mining —Text Mining - Text Data Analysis and Information Retrieval, Text mining approaches - Web Mining- Web Content Mining, Web Structure Mining, Web Usage Mining. CRM Applications and Data mining - Practical Data Mining Tools: Weka, RapidMiner.	9
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Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Assignment/ Microproject	Internal Examination-1 (Written)	Internal Examination- 2 (Written)	Total
5	15	10	10	40

End Semester Examination Marks (ESE):

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• 2 questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. • Each question carries 9 marks. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Understand the basic concepts in data mining and the data preprocessing.	K2
CO2	Describe the architecture, components, and design of data warehouses, including ETL processes and schema designs.	K2
CO3	Apply various data mining techniques such as classification, clustering, and association rule mining to real-world datasets.	K3
CO4	Demonstrate advanced data mining topics and practical applications, demonstrating the ability to integrate and apply knowledge in a real-world context.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	2			-	-	-	-	-	-	-	
CO2	2	2	1			-	-	-	-	-	-	1
CO3	2	2	2	1		-	-	-	-	-	-	1
CO4	2	2	2	1	2	-	-	-	-	-	-	1

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Data Mining Concepts and Techniques	Jaiwei Han, MichelineKamber	Elsevier	2nd Edition, 2006.
2	Data Warehousing, Data Mining & OLAP	AlexBerson and Stephen J. Smith,	TataMcGraw Hill	Edition, Tenth Reprint 2007.

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Data Warehousing Fundamentals: A Comprehensive Guide for IT Professionals	Pualraj Ponnaiah	Wiley	2nd edition, 2001
2	Data Mining: Introductory and Advanced Topics	Dunham M H	Pearson Education	New Delhi, 2003.
3	Data Mining; Practical Machine Learning Tools and Techniques	Ian Witten, Eibe Frank	Morgan Kaufmann	3rd edition, 2011
4	Introduction to Data Mining	Pang-Ning Tan and Michael Steinbach	Addison Wesley	2nd edition, 2006
5	Data Mining and Warehousing	M Sudeep Elayidom	Cengage Learning India Pvt. Ltd	1stEdition, 2015

Video Links (NPTEL, SWAYAM...)	
No.	Link ID
1	https://onlinecourses.nptel.ac.in/noc21_cs06/preview

SEMESTER S5

FORMAL LANGUAGES AND AUTOMATA THEORY

Course Code	PEITT525	CIE Marks	40
Teaching Hours/Week (L: T:P: R)	3:0:0:0	ESE Marks	60
Credits	5/3	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	GAMAT301: Mathematics for Information Science-3	Course Type	Theory

Course Objectives:

1. To design various computational models useful for solving problems.
2. To understand the relationship among digital computer, algorithm and Turing machine.
3. To discuss the Chomsky classification of formal languages with discussion on grammar and automata for regular, context-free, context-sensitive, and unrestricted languages.
4. To verify whether a given problem is solvable or tractable.

SYLLABUS

Module No.	Syllabus Description	Contact Hours
1	Finite Automata: Need for mathematical modelling, Symbols, Alphabets, Strings, Languages, Formal definition of a finite automaton, Deterministic Finite Automata (DFA), Regular languages, Formal definition of a nondeterministic finite automaton, Equivalence of NFAs and DFAs (Proof not expected) - Subset Construction. Minimization of DFA NFA with epsilon transitions, Eliminating epsilon transitions, Moore and Mealy machines .	9
2	Regular Expressions: Formal definition of a regular expression, Building Regular Expressions, Equivalence with finite automata (Proof not expected) - Converting FA to Regular Expressions, Converting Regular Expressions to FA, Regular languages. Properties of Regular Languages: Closure and Decision Properties of Regular Languages, Pumping Lemma for Regular Languages, Myhill-Nerode Theorem(MNT) without proof, Applications of MNT. Context-Free Grammars and Applications: Formal definition of a	9

	context-free grammar, Designing context-free grammars, Parse Trees, Ambiguous Grammars, CYK-Algorithm.	
3	Pushdown Automata: Formal definition of a pushdown automaton (PDA), Design of Deterministic PDA and Nondeterministic PDA. Simplification of Context-Free Languages: Chomsky normal form, Greibach normal form. Properties of Context-Free Languages: The Pumping Lemma for Context-Free Languages, Closure and Decision Properties of Context-Free Languages. Linear Bounded Automata	9
4	Turing Machines: Definition of a Turing machine (TM), Design of Turing machines – TM as language acceptors, TM as computers of functions, Variants of Turing Machines, Recursive and recursively enumerable languages Chomsky hierarchy. Computability: Universal Machine, Church Turing thesis, Decidable and Undecidable Problems, Halting problem of TM.	9

Course Assessment Method
(CIE: 40 marks, ESE: 60 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	Internal Examination (Written)	Evaluate Level Assessment	Analyze Level Assessment	Total
5	15	10	10	40

Evaluate and Analyze Level Assessment [20 Marks]

Design and implement a tool that handles various formal language concepts. Your tool should integrate functionalities for the following tasks:

- 1. Finite Automata Simulation:** Simulate both deterministic finite automata (DFA) and non-deterministic finite automata (NFA). Your tool should allow users to input an automaton and a string and should output whether the string is accepted by the automaton.
- 2. Regular Expression Parsing:** Parse regular expressions to construct an equivalent NFA or DFA. The tool should allow users to input a regular expression and a string, and determine if the string matches the regular expression.
- 3. Context-Free Grammar Checking:** Verify if a string can be generated by a given context-free grammar (CFG). Users should be able to input a CFG and a string, and your tool should check if the string can be derived from the CFG.

4. Turing Machine Simulation: Implement a Turing machine simulator where users can define the machine's states, transitions, input alphabet, and initial configuration. The tool should simulate the Turing machine's operation on a given input tape.
5. Automata Conversion: Implement algorithms to convert between different types of automata, such as converting an NFA to a DFA or converting a regular expression to an NFA.
6. Use JFLAP to evaluate and demonstrate your tool's functionalities.

Criteria for evaluation:

1. Finite Automata Simulation (K4- 4 points)

- a. **Application (K3- 2 points):** Ability to simulate DFA and NFA correctly. The tool should accept an automaton and a string, and accurately determine whether the string is accepted.
- b. **Analysis (K4- 2 points):** Proper handling of different types of automata inputs (DFA/NFA) and edge cases. Includes correct processing of input strings and states.

2. Regular Expression Parsing (K4- 4 points)

- a. **Application (K3-2 points):** Correct parsing of regular expressions to construct equivalent NFA or DFA. The tool should accurately build the automaton and match strings against the regular expression.
- b. **Analysis (K4- 2 points):** Handling of complex regular expressions and accurate matching results. Proper conversion and processing of regular expressions.

3. Context-Free Grammar Checking (K4- 4 points)

- a. **Application (K3- 2 points):** Correctly verify if a string can be generated by a given CFG. The tool should accept a CFG and a string and check if the string can be derived from the CFG.
- b. **Analysis (K4- 2 points):** Handling of different CFGs and strings, including edge cases and complex grammars.

4. Turing Machine Simulation (K4- 4 points)

- a. **Application (K3- 2 points):** Functional simulation of a Turing machine based on user-defined states, transitions, and input. Accurate simulation of operations on an input tape.
- b. **Analysis (K4- 2 points):** Proper handling of various Turing machine configurations and input scenarios. Effective simulation of state transitions and tape operations.

5. Automata Conversion (K4- 2 points)

- a. **Application (K3- 1 point):** Correct implementation of algorithms to convert between different types of automata (e.g., NFA to DFA).
- b. **Analysis (K4- 1 point):** Accurate conversion processes and handling of edge cases. Effective demonstration of the conversion algorithms in practice.

6. Integration with JFLAP (K5- 2 points)

- a. **Evaluation (1 point):** Effective use of JFLAP for visual representation and evaluation of the tool's functionalities.
- b. **Presentation (1 point):** Clear and effective demonstration of the tool using JFLAP, including accurate representation of results.

Scoring:

1. **Accomplished (4 points):** Exceptional analysis, clear implementation, and depth of understanding.
2. **Competent (3 points):** Solid performance with minor areas for improvement.
3. **Developing (2 points):** Adequate effort but lacks depth or clarity.
4. **Minimal (1 point):** Incomplete or significantly flawed.

End Semester Examination Marks (ESE):

In Part A, all questions need to be answered and in Part B, each student can choose any one full question out of two questions

Part A	Part B	Total
• 2 Questions from each module. • Total of 8 Questions, each carrying 3 marks (8x3 =24marks)	• Each question carries 9 marks. • Two questions will be given from each module, out of which 1 question should be answered. • Each question can have a maximum of 3 sub divisions. (4x9 = 36 marks)	60

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Explain the fundamental concepts in automata theory and formal languages.	K2
CO2	Design finite state automata, regular grammar and regular expression for regular languages.	K3
CO3	Design push-down automata and context-free grammar representations for context-free languages.	K3
CO4	Design Turing Machines to accept recursive and recursively enumerable languages.	K3
CO5	Analyse the limitations of computational models and understand decidability and undecidability of problems.	K4

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	2	1	1	-	-	-	-	-	-	-	-	1
CO2	3	3	3	2	-	-	-	-	-	-	-	1
CO3	3	3	3	2	-	-	-	-	-	-	-	1
CO4	3	3	3	2	-	-	-	-	-	-	-	1
CO5	3	3	1	1	-	-	-	-	-	-	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to the Theory of Computation	Michael Sipser	Cengage India Private Limited	Third Edition, 2014
2	Introduction to Automata Theory Languages and Computation	Hopcroft, Motwani, and Ullman	Pearson Publishers	Third Edition, 2006
3	An Introduction to Formal Languages and Automata	Peter Linz and Susan H. Rodger	Jones and Bartlett Publishers, Inc	Seventh Edition, ,2022

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Automata and Computability	Dexter C. Kozen	Springer	2007
2	Introduction to Languages and the Theory of Computation	John Martin	McGraw-Hill Education	Third Edition, 2003

Video Links (NPTEL, SWAYAM...)	
No.	Link ID
1	https://archive.nptel.ac.in/courses/106/104/106104148/
2	https://nptel.ac.in/courses/106106049

SEMESTER S5 MACHINE LEARNING LAB

Course Code	PCITL507	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	None	Course Type	LAB

Course Objectives:

1. By the end of this course, students will be able to implement and evaluate multiple machine learning models, including Multiple Linear Regression, Logistic Regression, Decision Trees (ID3 & CART), Naïve Bayes' Classifier, and Support Vector Machines, using practical datasets such as house prices, Titanic survival data, and spam emails.
2. Gain hands-on experience in applying advanced machine learning techniques to various tasks, including character recognition using Multilayer Perceptron, clustering with K-Means and DBScan, sentiment analysis with Random Forest, Adaboost, and XGBoost, Q-Learning for game strategies, and image classification using Convolutional Neural Networks (CNNs).

Experiment No.	Experiment
1	Study of Python Libraries for ML applications such as Pandas and Matplotlib.
2	Implement Multiple Linear Regression for House Price Prediction.
3	Implement Logistic Regression for Titanic Survival Prediction without using Libraries.
4	Implement Decision trees – ID3 & CART.
5	Implement Naïve Bayes' Classifier for Diagnosis of Heart Patients.
6	Implement Support Vector Machines to Filter Spam Mails.
7	Implement Multilayer Perceptron for Character Recognition using MNIST Dataset
8	Implement k-Nearest Neighbor without using libraries for IRIS Dataset
9	Implement K-Means Clustering & DBScan.
10	Implement Sentiment Analysis using Random Forest, Adaboost, XGboost
11	Implement Q-Learning for a Simple Game.

12	Implement CNN for Image Classification.
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Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)

Continuous Internal Evaluation Marks (CIE):

Attendance	<i>Preparation/Pre-Lab Work, experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)</i>	<i>Internal Exam</i>	Total
5	20	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

- **Submission of Record:** Students shall be allowed for the end semester examination only upon submitting the duly certified record.
- **Endorsement by External Examiner:** The external examiner shall endorse the record.

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Demonstrate proficiency in Python Libraries for Machine Learning, Python libraries such as Pandas and Matplotlib to preprocess data, visualize datasets, and gain insights essential for machine learning applications.	K3
CO2	Implement various machine learning algorithms, including Multiple Linear Regression, Logistic Regression, Decision Trees (ID3 & CART), and Naïve Bayes' Classifier, and will be able to assess their performance on different datasets.	K3

CO3	Build and apply advanced models such as Multilayer Perceptron for character recognition, Support Vector Machines for spam filtering, and Convolutional Neural Networks for image classification, as well as implement clustering techniques like K-Means and DBScan.	K3
CO4	Implement Q-Learning for simple game scenarios and perform sentiment analysis using ensemble methods such as Random Forest, Adaboost, and XGboost, demonstrating a comprehensive understanding of both supervised and reinforcement learning techniques.	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12
CO1	3	3	2	2	2	2	-	-	2	-	-	2
CO2	3	3	2	2	2	2	-	-	2	-	-	2
CO3	3	3	2	2	2	2	-	-	2	-	-	2
CO4	3	3	2	2	2	2	-	-	2	-	-	2

Note: 1: Slight (Low), 2: Moderate (Medium), 3: Substantial (High), -: No Correlation

Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Python Machine Learning	Sebastian Raschka (Author), Vahid Mirjalili (Author)	Packt	2 nd Edition, 2019
2	Machine Learning	Tom M Mitchell	McGraw Hill Education	First Edition, 2013
3	Introduction to Machine Learning with Python	Andreas C. Müller, Sarah Guido	O'Reilly Media, Inc.	2016

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow,	Aurélien Géron	O'Reilly Media, Inc.	2 nd Edition, 2019

Video Links (NPTEL, SWAYAM...)	
No.	Link ID
1	https://cse.iitkgp.ac.in/~pabitra/course/ml/ml.html

Continuous Assessment (20 Marks)

1. Preparation and Pre-Lab Work (5 Marks)

- Pre-Lab Assignments: Assessment of pre-lab assignments or quizzes that test understanding of the upcoming experiment.
- Understanding of Theory: Evaluation based on students' preparation and understanding of the theoretical background related to the experiments.

2. Conduct of Experiments (5 Marks)

- Procedure and Execution: Adherence to correct procedures, accurate execution of experiments, and following safety protocols.
- Skill Proficiency: Proficiency in handling equipment, accuracy in observations, and troubleshooting skills during the experiments.
- Teamwork: Collaboration and participation in group experiments.

3. Lab Reports and Record Keeping (5 Marks)

- Quality of Reports: Clarity, completeness and accuracy of lab reports. Proper documentation of experiments, data analysis and conclusions.
- Timely Submission: Adhering to deadlines for submitting lab reports/rough record and maintaining a well-organized fair record.

4. Viva Voce (5 Marks)

- Oral Examination: Ability to explain the experiment, results and underlying principles during a viva voce session.

***Final Marks Averaging:** The final marks for preparation, conduct of experiments, viva, and record are the average of all the specified experiments in the syllabus.*

Evaluation Pattern for End Semester Examination (50 Marks)

1. Procedure/Preliminary Work/Design/Algorithm (10 Marks)

- Procedure Understanding and Description: Clarity in explaining the procedure and understanding each step involved.
- Preliminary Work and Planning: Thoroughness in planning and organizing materials/equipment.
- Algorithm Development: Correctness and efficiency of the algorithm related to the experiment.
- Creativity and logic in algorithm or experimental design.

2. Conduct of Experiment/Execution of Work/Programming (15 Marks)

- Setup and Execution: Proper setup and accurate execution of the experiment or programming task.

3. Result with Valid Inference/Quality of Output (10 Marks)

- Accuracy of Results: Precision and correctness of the obtained results.

- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. Record (5 Marks)

- Completeness, clarity, and accuracy of the lab record submitted.

SEMESTER S5

ALGORITHM ANALYSIS AND DESIGN LAB

Course Code	PCITL508	CIE Marks	50
Teaching Hours/Week (L: T:P: R)	0:0:3:0	ESE Marks	50
Credits	2	Exam Hours	2 Hrs. 30 Min.
Prerequisites (if any)	PCITL308: Datastructures Lab	Course Type	Practical

Course Objectives:

1. To equip students to develop proficiency in Algorithmic Techniques and Data Structures
2. Enable students to design and implement algorithms for solving complex computational problems using C programming language.

SYLLABUS

Experiment No.	Experiment
1	Implement BFS on an undirected graph.
2	Implement DFS on a directed graph.
3	Implement the program to find the Strongly Connected Components of a directed graph.
4	Implement a Program to find the Topological Sorting of a Directed Graph.
5	Implement the divide and conquer algorithm to find Minimum and Maximum in an array of elements.
6	Implement Merge Sort
7	Implement Quick Sort
8	Implement Greedy algorithm to find the solution to fractional Knapsack Problem.
9	Implement Kruskal's algorithm to find Minimum Spanning Tree.

10	Implement Prim's Algorithm to find Minimum Spanning Tree.
11	Implement Dynamic Programming algorithm to find the solution to Travelling Salesman Problem.
12	Implement Dynamic Programming algorithm to find the solution to 0/1 Knapsack Problem.
13.	Implement Backtracking algorithm to find the solution to N-Queen's problem.
14.	Implement KMP String Matching Algorithm.
15.	Implement Rabin Karp String Matching Algorithm.

**Course Assessment Method
(CIE: 50 marks, ESE: 50 marks)**

Continuous Internal Evaluation Marks (CIE):

Attendance	Preparation/Pre-Lab Work, experiments, Viva and Timely completion of Lab Reports / Record (Continuous Assessment)	Internal Examination	Total
10	20	20	50

End Semester Examination Marks (ESE):

Procedure/ Preparatory work/Design/ Algorithm	Conduct of experiment/ Execution of work/ troubleshooting/ Programming	Result with valid inference/ Quality of Output	Viva voce	Record	Total
10	15	10	10	5	50

- **Submission of Record:** Students shall be allowed for the end semester examination only upon submitting the duly certified record.
- **Endorsement by External Examiner:** The external examiner shall endorse the record.

Course Outcomes (COs)

At the end of the course students should be able to:

Course Outcome		Bloom's Knowledge Level (KL)
CO1	Implement Graph Traversals and their applications.	K3
CO2	Implement Divide and Conquer and Greedy Technique.	K3
CO3	Implement Dynamic Programming and Backtracking Techniques	K3
CO4	Implement String Matching Algorithms	K3

Note: K1- Remember, K2- Understand, K3- Apply, K4- Analyse, K5- Evaluate, K6- Create

CO-PO Mapping Table (Mapping of Course Outcomes to Program Outcomes)

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Text Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	Introduction to Algorithms	T.H.Cormen, C.E.Leiserson, R.L.Rivest, C. Stein	Prentice-Hall India	2nd Edition (2001)
2	Fundamentals of Computer Algorithms	Ellis Horowitz, Sartaj Sahni, Sanguthevar Rajasekaran	Orient Longman Universities Press	2nd Edition (2008)
3	Computer Algorithms, Introduction to Design and Analysis	Sara Baase and Allen Van Gelder	Pearson Education	3rd Edition (2009)

Reference Books				
Sl. No	Title of the Book	Name of the Author/s	Name of the Publisher	Edition and Year
1	A Practical Guide : to Analysis and Design of Algorithms	Dr. Deepshikha Agarwal	Notion Press	January 2021
2	Programming In Ansi C	E Balagurusamy	McGraw Hill Education	8 th edition, March 2019

Video Links (NPTEL, SWAYAM...)	
No.	Link ID
1	https://onlinecourses.nptel.ac.in/noc19_cs47/preview

Continuous Assessment (20 Marks)

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3. Result with Valid Inference/Quality of Output (10 Marks)

- Accuracy of Results: Precision and correctness of the obtained results.
- Analysis and Interpretation: Validity of inferences drawn from the experiment or quality of program output.

4. Viva Voce (10 Marks)

- Ability to explain the experiment, procedure results and answer related questions
- Proficiency in answering questions related to theoretical and practical aspects of the subject.

5. Record (5 Marks)

- Completeness, clarity, and accuracy of the lab record submitted

SEMESTER 6

INFORMATION TECHNOLOGY